Review: Solving one-step inequalities

1 Determine whether the following values satisfy the inequality $x + 3 > 5$.

a $x = 1$ **b** $x = 3$ **c** $x = -9$ **d** $x = 2$

2 Solve the following inequalities.

a $x + 5 \geq 11$ **b** $9 > y + 4$ **c** $n + 8 \leq 7$ **d** $m - 5 > 9$

e $7 \leq x - 3$ **f** $x - 8 < 11$ **g** $x - 8 \leq -4$ **h** $9y > 81$

i $10x < 90$ **j** $56 \leq 8p$ **k** $3x \leq -18$ **l** $\frac{x}{3} \geq 6$

m $\frac{n}{7} > 8$ **n** $8 \geq \frac{x}{3}$ **o** $-4x < 32$ **p** $-4y \geq -16$

q $-5x \leq -40$ **r** $-24 < -8m$ **s** $\frac{a}{6} < 11$ **t** $\frac{x}{-9} \geq 5$

u $\frac{x}{-7} < 2$

3 For each of the following:

i Write down the inequality described by the statement.

ii Solve for the value of x.

a "Seven more than the value of x is at least nine".

b "Half of x is no more than four".

c "Eight is greater than the result of taking seven away from x".

d "Negative eight groups of x is less than three".

4 For each of the following:

i Solve the inequality.

ii Plot the solution to the inequality on a number line.

a $8 + x \leq 8$ **b** $x + 1 < 4$

c $x - 4 < 1$ **d** $3 + x < 2$

e $-8 < x - 4$ **f** $-2 + x \leq 4$

g $2 + x \geq 7$ **h** $8 < 10 + x$

i $-5 + x < -7$ **j** $x - 2 \geq -3$

k $-4 < -3 + x$ **l** $6x \leq 48$

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m $36 \leq 9x$

n $2x > -4$

o $7x < -35$

p $-5x \leq 15$

q $-3x > -21$

r $72 > -8x$

s $\frac{x}{2} \geq 7$

t $\frac{x}{6} < -2$

u $-6 \geq \frac{x}{3}$

v $\frac{x}{-8} \leq -2$

w $\frac{x}{-5} > 2$

x $\frac{x}{-7} < 2$

y $-2 \leq \frac{x}{-7}$

z $\frac{x}{3} > \frac{4}{3}$

5 For each of the following:

i Solve the inequality.

ii Plot the solution to the inequality on a number line.

a $x + \frac{7}{4} \geq \frac{9}{4}$

b $-4.5x \geq -18$

c $x + 2.5 \leq 7.5$

d $x + 3.5 > 8$

e $x - 2.6 \leq 2.4$

f $-4.5 + x \geq 4$

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